

EXPERENTIAL LEARNING LAB1

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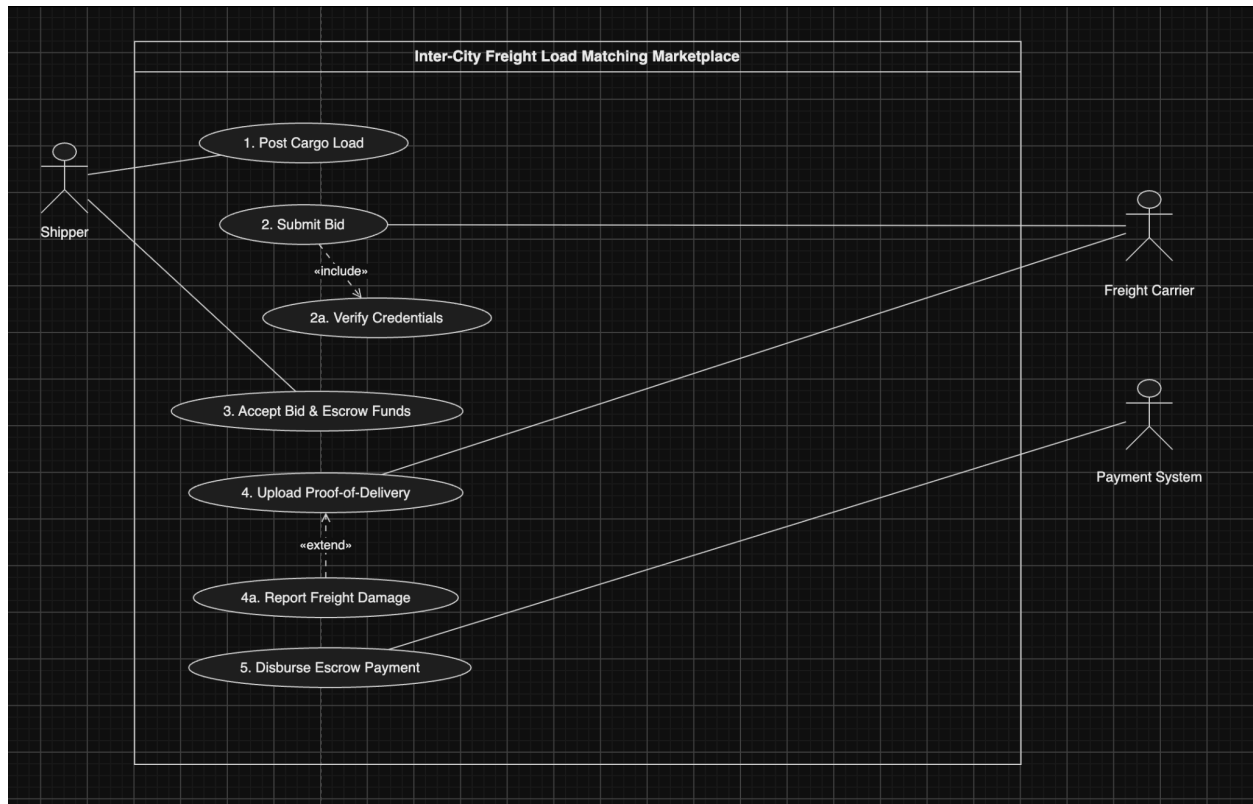
Title- Inter-City Freight Load Matching Marketplace(28)

1) Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow verified trucking carriers to submit competitive bids on posted freight shipments within a specified auction timeframe.	High	Pass: Carrier bid recorded and shipper notified. Fail: Unverified carrier submits bid without valid permits.	Enables the core bidding marketplace mechanics.
FR-002	Functional	The system shall allow commercial shippers to post cargo loads with details including weight, dimensions, and delivery deadlines.	High	Pass: Load is posted to the active marketplace board. Fail: System rejects post missing mandatory dimension details.	Shippers need a standardized way to request freight services.
FR-003	Functional	The system shall allow shippers to evaluate bids and select a winning freight carrier.	High	Pass: Winning carrier is assigned to the load and other bidders are notified of rejection.	Necessary to close the auction and initiate the delivery contract.
FR-004	Functional	The system shall hold milestone payments in escrow upon the shipper accepting a carrier's bid.	High	Pass: Funds are verified and locked in escrow; carrier is notified to begin transit.	Protects the carrier by guaranteeing funds are available before transit.

FR-005	Functional	The system shall allow carriers to submit an electronic proof-of-delivery sign-off upon dropping off the cargo.	High	Pass: Document is uploaded, scanned for validity, and flagged for payment release.	Required to trigger the final payment process.
NFR-001	Performance & Security	Milestone payments held in escrow shall disburse automatically within 60 seconds of verified electronic proof-of-delivery submission.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures fast, reliable payment for carriers, building trust in the B2B platform.
NFR-002	Usability & Availability	The marketplace platform shall maintain 99.9% uptime during normal business hours to ensure uninterrupted bidding.	Medium	Pass: Server logs indicate less than 8.7 hours of downtime per year.	The logistics industry operates continuously; downtime causes lost revenue and delayed shipments.

2) UML Diagram



3) Use-Case Flow Specification (Main Success & Alternate Flow)

Use Case Name: Submit Freight Bid **Primary Actor:** Freight Carrier

Preconditions:

- The Freight Carrier must be logged into their registered platform account.
- The Shipper must have an active, unexpired cargo load posted on the marketplace.

Postconditions:

- A valid, competitive bid is recorded in the database and linked to the specific cargo load.
- The Shipper is successfully notified of the new bid.

Main Success Scenario (Happy Path):

1. The Carrier browses the active marketplace board and selects a cargo load they wish to transport.
2. The Carrier enters their proposed monetary bid and estimated transit timeframe, then clicks "Submit".

3. The system executes the included use case "Verify Credentials" to check the backend database for the Carrier's active trucking permits and insurance validity.
4. The system validates that the bid meets the auction parameters (e.g., submitted before the deadline).
5. The system successfully records the bid.
6. The system triggers an automated notification to the Shipper indicating a new bid has been received for review.

Alternate Flow (Failed Credential Verification): This alternate flow occurs at Step 3 of the Main Success Scenario.

1. The system checks the database and discovers that the Carrier's insurance or trucking permits have expired.
2. The system immediately rejects the bid attempt to prevent an unauthorized contract.
3. The system displays an error prompt to the Carrier, explaining that their account is restricted due to expired documentation.
4. The Carrier is provided a link to the "Document Upload" portal to renew their credentials.
5. The use case terminates without recording the bid

4) Identified Actors and Core Use Cases

Identified Actors:

- **Shipper (Primary Actor):** The commercial entity that needs freight moved. They initiate the primary business flow by posting cargo loads and ultimately choosing who to hire.
- **Freight Carrier (Primary Actor):** The trucking company or driver utilizing the platform to find work. They interact with the system to bid on active loads and confirm deliveries.
- **Payment Gateway/System (Secondary Actor):** A background system (or external API) that securely holds funds in escrow and automatically disburses them once delivery conditions are met.

Core Use Cases:

1. **Post Cargo Load:** The Shipper enters freight details (weight, dimensions, delivery deadlines) to the marketplace board.
2. **Submit Bid:** The Carrier offers a competitive price and timeframe to transport the load.
3. **Verify Credentials <<include>>:** A mandatory background check triggered during the bidding process to ensure the Carrier is legally allowed to operate.
4. **Accept Bid & Escrow Funds:** The Shipper selects a winning bid, which triggers the system to lock the required payment in a secure hold.
5. **Upload Proof-of-Delivery:** The Carrier submits an electronic document or signature confirming they have dropped off the cargo.
6. **Report Freight Damage <<extend>>:** An optional extension of the delivery upload process, triggered *only* if the cargo was damaged during transit.

7. **Disburse Escrow Payment:** The Payment System automatically releases the held funds to the Carrier upon verified delivery.